

$$\text{PAPER_1599} - \text{Carbon Steel Density} = \rho_{\text{steel}} = D_{\text{crit}}^2 \cdot SO_5 + SO_5^3 + SO_5 \cdot N_{\text{CH}} = 6760 + 1000 + 90 = 7850 \text{ kg/m}^3$$

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Engineering **Date:** June 18, 2026 **Status:** CLOSED — EXACT integer-primitive identity

Observation

PAPER_1209Y_S568 (Engineering Unified Proof Set) lists **Carbon Steel Density** with the integer-primitive expression:

$$\rho_{\text{steel}} = D_{\text{crit}}^2 \cdot SO_5 + SO_5^3 + SO_5 \cdot N_{\text{CH}} = 6760 + 1000 + 90 = 7850 \text{ kg/m}^3$$

Residual: **EXACT**.

UQFF Closed Identity

$$\rho_{\text{steel}} = D_{\text{crit}}^2 \cdot SO_5 + SO_5^3 + SO_5 \cdot N_{\text{CH}} = 6760 + 1000 + 90 = 7850 \text{ kg/m}^3 \quad \text{EXACT}$$

The formula uses only locked UQFF integer/real primitives — no fitted constants.

NOT REPLACEMENT

Empirical engineering measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1209Y_S568
- Related: PAPER_1494-1595 (other session-2026-06-18 closures)
- Calculator dispatch: `calculate_paradox({"paradox": "carbon_steel_density_7850"})`

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